

Signaling games

Signaling game

- Two players, a sender and a receiver.
- After observing her type, sender sends a message or signal to receiver.
- The receiver observes the signal and responds with an action.
- Players' payoffs depend on type, signal, and response.
- A probability distribution, a move by nature, determines the sender's type at the beginning of the game. Interpretation: receiver's a priori beliefs.

Applications

- Information economics: Akerlof (1970), Spence (1973), and Rothschild and Stiglitz (1976) among many others.
- Information transmission with cheap talk, Crawford and Sobel (1982), Kono and Kandori (2019).

Other: Chakraborty and Harbaugh (2010), Kamenica and Gentzkow (2011).

- Game theory. Forward induction ideas in signaling games, Cho and Kreps (1987).

Other applications

- Advertising as signal of product quality, education as a signal of productivity, Milgrom and Roberts (1986), Bagwell and Riordan (1991).
- Finance. Firm's debt-equity ratio and the Modigliani-Miller theorem, Leland and Pyle (1977), Myers and Majluf (1984), Ross (1977).
- Marketing, accounting, and many more, Bagwell and Ramey (1994).
- Political sciences, Martinelli and Matsui (2002).

Signaling game

Definition 1 (Signaling game) A signaling game is a 2-person, extensive-form game $\Gamma = [(T, P), A_1, A_2, U_1, U_2]$ where for $i = 1, 2$

1. T is a space with a probability distribution P .
 2. A_i is player i 's action space.
 3. $U_i : T \times A_1 \times A_2 \rightarrow \mathbb{R}$ is i 's payoff function.
 4. Player 1 observes her type $t \in T$ and chooses an action $a_1 \in A_1$.
Player 2 does not observe t . Player 2 observes a_1 , makes an inference about player 1's type and chooses $a_2 \in A_2$.
- Implicitly it has been assumed that T and A_i are discrete. Many applications of interest use a continuum of types or actions.

Definitions: Type of equilibria

Usually defined for pure strategies.

Definition 2 Let (σ_1, σ_2) be an equilibrium of the signaling game Γ .

- If $t, t' \in T, t \neq t' \implies \text{supp} [\sigma_1(t)] \cap \text{supp} [\sigma_1(t')] = \emptyset$, then (σ_1, σ_2) is a separating equilibrium.
- If $t, t' \in T \implies \sigma_1(t) = \sigma_1(t')$, then (σ_1, σ_2) is a pooling equilibrium.

Example: separating equilibrium

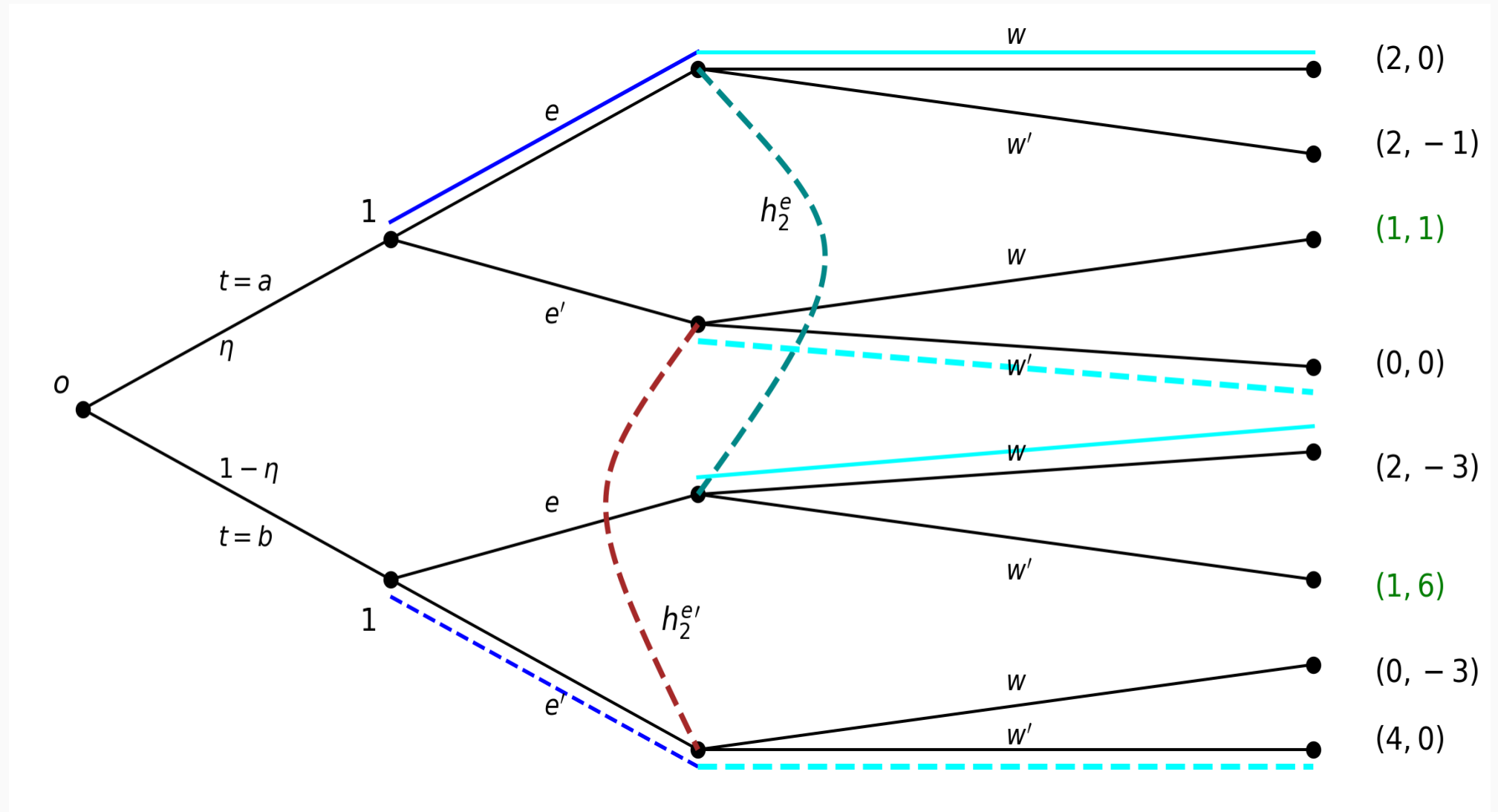


Figure 1

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- Separating (*blue*).
 - $s_1(h_1) = e, s_1(h'_1) = e'$
 - $s_2(h_2^e) = w, s_2(h_2^{e'}) = w'$
 - Find prior probabilities (i.e. the value of η) and the accompanying beliefs that make the strategy profile described above a WPBE.
 - Is the strategy profile described part of a Sequential equilibrium?

Example: pooling equilibrium

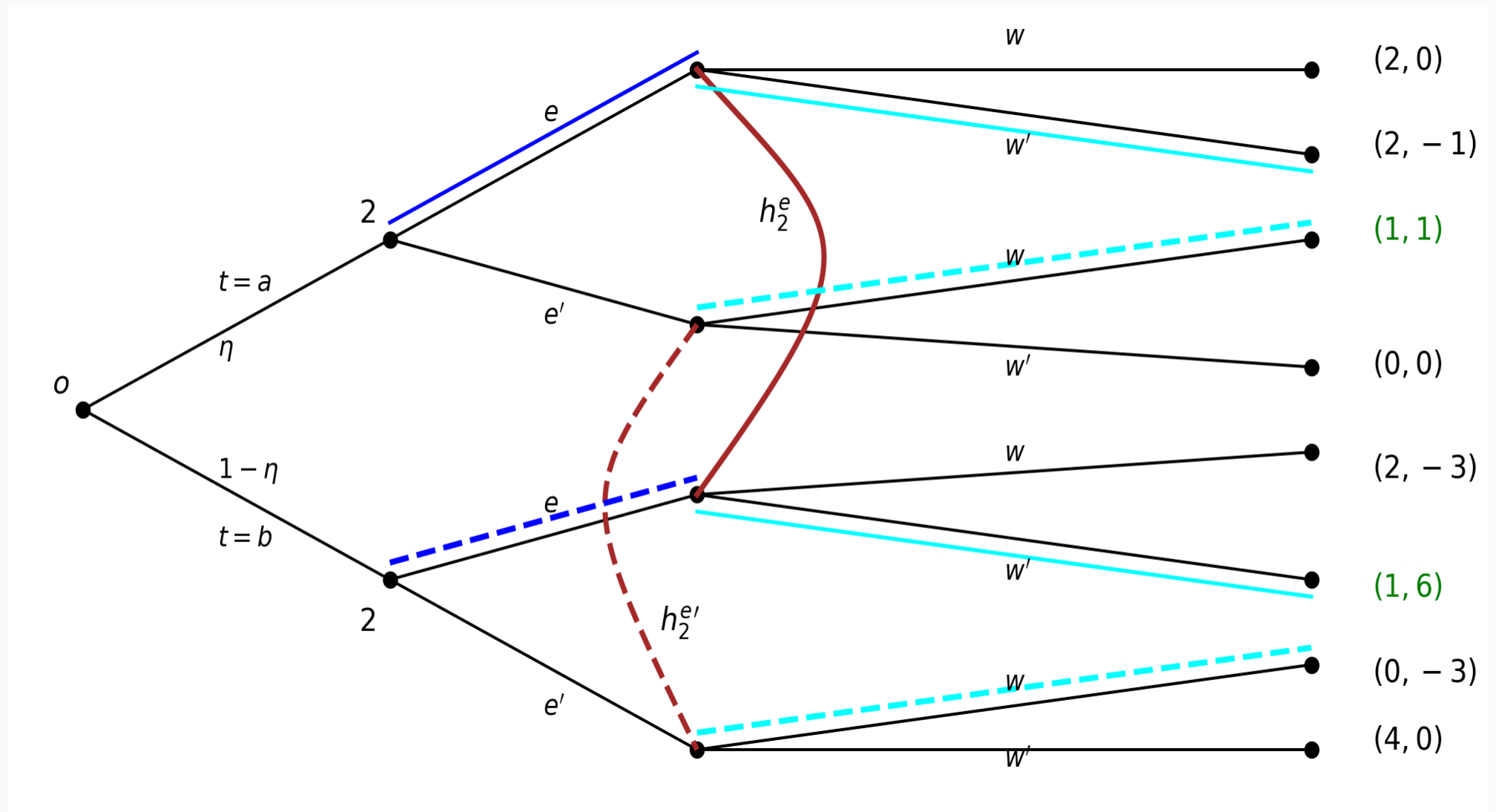


Figure 2

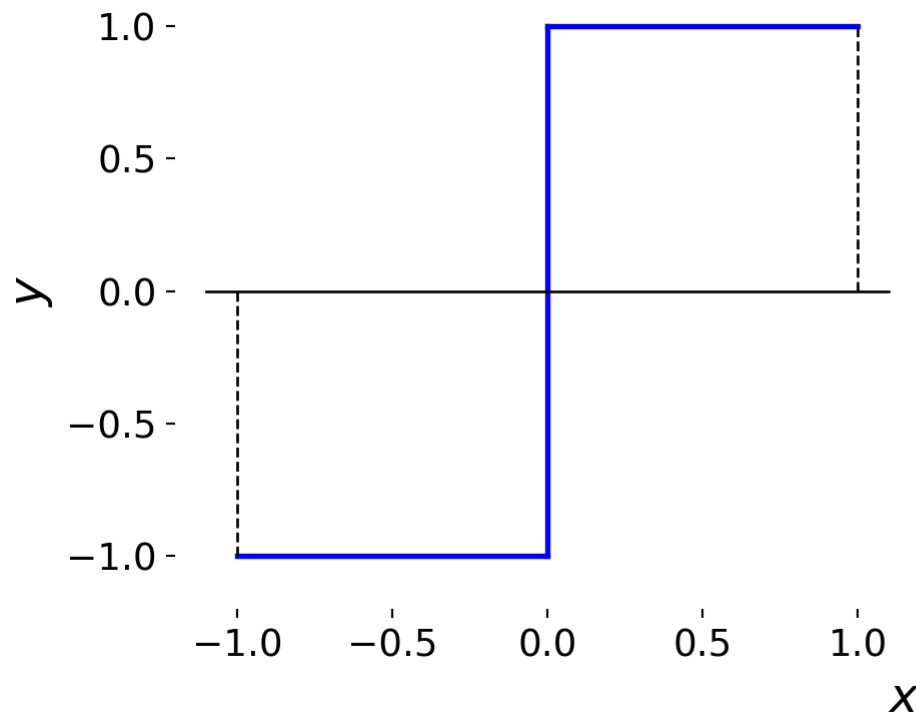
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- Pooling
 - $s_1(h_1) = s_1(h'_1) = e$
 - $s_2(h_2^e) = w', s_2(h_2^{e'}) = w$
 - Find prior probabilities (i.e. the value of η) and the accompanying beliefs that make the strategy profile described above a WPBE.
 - Is the strategy profile described part of a Sequential Equilibrium?

Non-existence infinite signaling game

- Two players, $i = 1, 2$.
- Player 1 observes his type $t \in T = \{-1, 1\}$ and sends a signal $x \in X = [-1, 1]$.
- Player 2 observes x and responds by selecting $y \in Y = [-1, 1]$.
- t follows the distribution $\rho(\{-1\}) = \rho(\{1\}) = 1/2$.
- Payoffs: $u_1(t, x, y) = -(t + x - y)^2$ and $u_2(x, y) = xy$.

See Manelli (1996).

Example



$$u_2(x, y) = xy$$

$$\tilde{y}(x) = -1 \text{ if } x < 0$$

$$\tilde{y}(x) = 1 \text{ if } x > 0$$

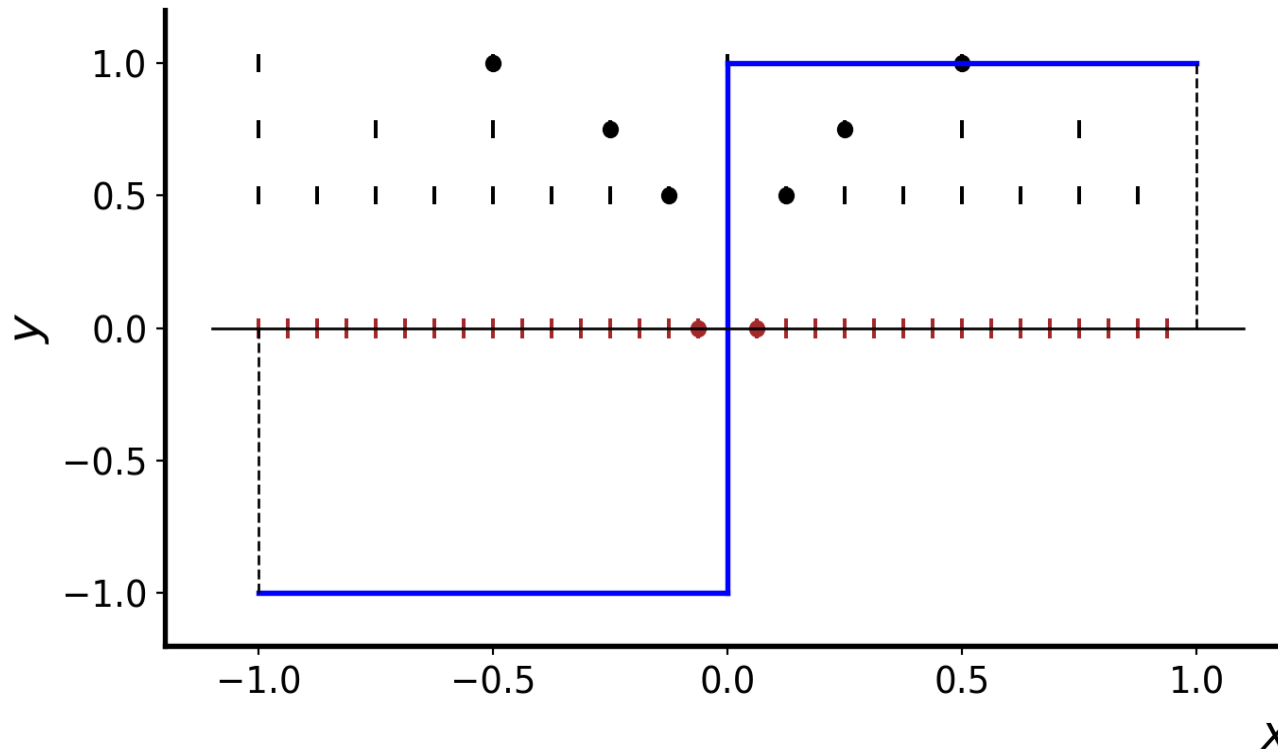
$$\tilde{y}(x) \in [-1, 1] \text{ if } x = 0$$

$$u_1(t, x, y) = -(t + x - y)^2$$

- $\tilde{y}(x)$ is 2's best response correspondence.
- $t = -1 \implies x \leq 0$, and as close to 0 as possible.
- $t > 1 \implies x \geq 0$, and as close to 0 as possible.

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- Finite versions of the game have equilibria.
 - Equilibrium outcomes are probability distributions over terminal nodes.
 - Any sequence of equilibrium outcomes of increasingly finer discretizations of the infinite game will converge (in a subsequence) to a limit distribution. (Weak convergence of probabilities.)
 - The limit distribution need not be feasible in the limit game, i.e., no pair of strategies of the limit game will generate that limit distribution. *If it is feasible, it is an equilibrium of the limit game.*

Example, finite approximation



$$u_2(x, y) = xy$$

$$\tilde{y}(x) = -1 \text{ if } x < 0$$

$$\tilde{y}(x) = 1 \text{ if } x > 0$$

$$\tilde{y}(x) \in [1, -1] \text{ if } x = 0$$

$$u_1(t, x, y) = -(t + x - y)^2$$

$$\tilde{y}(x) = \begin{cases} -1 & \text{if } x \leq 1/n \\ 1 & \text{if } x \geq 1/n \end{cases}$$

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- The finite game has a sequential equilibrium:

$$\hat{x}^n(t) = \begin{cases} -1/n & \text{if } t = -1 \\ 1/n & \text{if } t = 1 \end{cases}$$

- The outcome is $\hat{\lambda}^n$ on $T \times X^n \times Y^n$ given by
$$\hat{\lambda}^n(\{(-1, -1/n, -1)\}) = \hat{\lambda}^n(\{(1, 1/n, 1)\}) = 1/2.$$

- The outcome is $\hat{\lambda}^n$ on $T \times X^n \times Y^n$ given by

$$\hat{\lambda}^n(\{(-1, -1/n, -1)\}) = \hat{\lambda}^n(\{(1, 1/n, 1)\}) = 1/2.$$
- $\hat{\lambda}^n$ converges weakly to $\hat{\lambda}$ on $T \times X \times Y$ where

$$\hat{\lambda}(\{(-1, 0, -1)\}) = \hat{\lambda}(\{(1, 0, 1)\}) = 1/2.$$

- No strategy pair in the limit game can generate $\hat{\lambda}$:

Player 1 must play $\hat{x}(-1) = \hat{x}(1) = 0$.

Player 2, however, must respond to $x = 0$ with $y = -1$ when $t = -1$ and with $y = 1$ when $t = 1$.

- Coordination in the finite approximating games is lost in the limit game.

Correlation

- Some form of correlation must be added to the limit game to restore the lost coordination. Minimum necessary.
- Signaling games: one adds cheap talk.
- Standard Bayesian Game: public randomization device.
- Stage games (repeated games): public randomization devices per stage.

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